Complex Tori and Elliptic Curves

Why the Abelian Group Law Must Be the Chord–Tangent Law via the $\wp$-Embedding

Spring 2026

Detailed lecture notes based on the midterm and final exam themes: differential forms, residues, compact Riemann surfaces, divisors, Riemann–Roch, complex tori, and the Weierstrass $\wp$-function.

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1 Purpose of These Notes

These notes explain one central fact:

> The usual chord–tangent law on a smooth plane cubic is not an arbitrary geometric trick. It is exactly the abelian group law on the complex torus $\mathbb{C}/\Lambda$, transported to the cubic by the Weierstrass functions $\wp$ and $\wp'$.

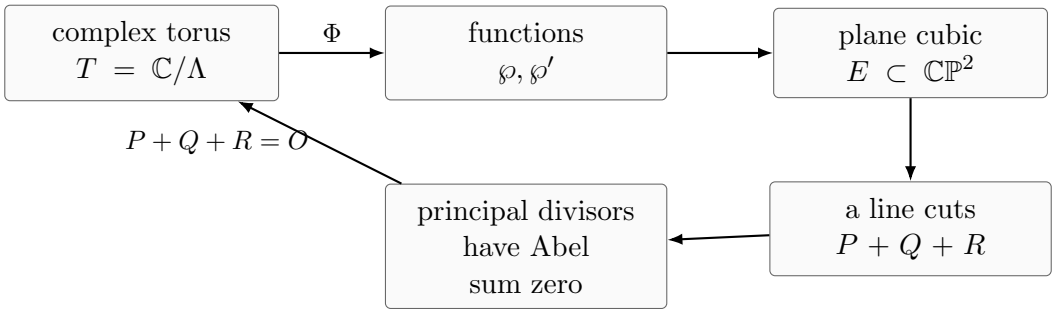
The slogan is:

$$\boxed{\text{addition on } \mathbb{C}/\Lambda \quad \Longleftrightarrow \quad \text{chord–tangent addition on } E}$$

where

$$E: \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

The reason this must be true is a chain of ideas that appeared throughout the midterm and final exams.



Main conclusion. If a line meets the cubic in three points P, Q, R, counted with intersection multiplicity, then those three points add to zero in the group law transported from the torus:

$$P + Q + R = O.$$

Solving for $P + Q$ gives

$$P + Q = -R.$$

Geometrically, this means: draw the line through P and Q, take the third intersection R, then reflect R across the x-axis.

2 Where the Midterm and Final Exam Material Enters

The argument is not just about elliptic curves. It uses many of the structural tools from the exams.

2.1 From the Midterm

1. **Differential forms and Stokes' theorem.** The midterm emphasized that integral theorems are best understood as statements about differential forms. The residue theorem on compact Riemann surfaces is another global Stokes-type statement.

2. **Cauchy–Green and residues.** The proof that residues are global obstructions is the same philosophy: local singular data must satisfy a global sum condition.

3. **The torus as a quotient.** The midterm computed geometric invariants of a real torus and a complex torus. In the present notes the complex torus

   $$T = \mathbb{C}/\Lambda$$

   is the source of the group law.

4. **Euler characteristic of a torus.** The midterm showed that the torus has Euler characteristic 0. For a compact Riemann surface,

   $$\chi = 2 - 2g,$$

   so the complex torus has genus $g = 1$. This is exactly the genus needed for elliptic curves.

2.2 From the Final

1. **Compact Riemann surfaces and meromorphic maps.** The final used the fact that nonconstant holomorphic maps between compact Riemann surfaces are finite branched coverings. The functions $\wp$ and $\wp'$ are meromorphic functions on a compact Riemann surface, namely $\mathbb{C}/\Lambda$.

2. **Principal divisors have degree zero.** The final proved this using the logarithmic derivative df/f. That is the first half of the Abel sum law.

3. **Riemann–Roch.** On a genus-one surface, Riemann–Roch predicts exactly the dimensions that force the functions $1, \wp, \wp'$ to be the natural coordinate functions.

4. **The torus and $\wp$.** The final problem on $\wp$ derived

   $$(\wp')^2 = 4\wp^3 - g_2\wp - g_3.$$

   This is precisely the equation of the plane cubic.

Exam-to-lecture bridge. The midterm supplies the geometric and differential-form intuition. The final supplies the compact Riemann surface tools. Together they explain why the algebraic-looking chord–tangent rule is actually forced by complex analysis.

3 The Complex Torus

3.1 Lattices

Definition. *A* lattice *in* $\mathbb{C}$ *is a subgroup*

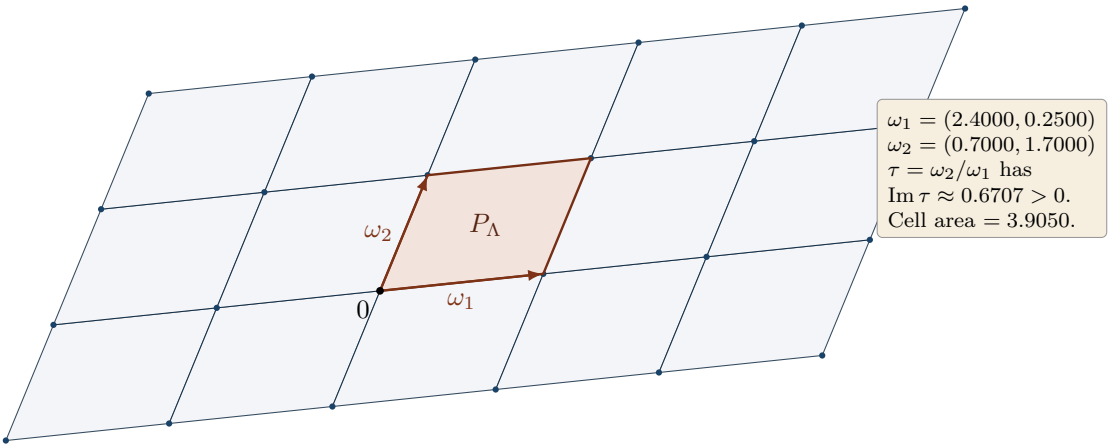
$$\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

where $\omega_1, \omega_2 \in \mathbb{C}$ *are linearly independent over* $\mathbb{R}$*. Equivalently,*

$$\frac{\omega_1}{\omega_2} \notin \mathbb{R}.$$

The condition of real-linear independence means that the two periods span a parallelogram of nonzero area in the complex plane.

$$P_\Lambda = \{s\omega_1 + t\omega_2 : 0 \le s, t \le 1\}.$$



3.2 The Quotient Surface

Define an equivalence relation on $\mathbb{C}$ by

$$z \sim w \quad \Longleftrightarrow \quad z - w \in \Lambda.$$

The quotient is

$$T = \mathbb{C}/\Lambda.$$

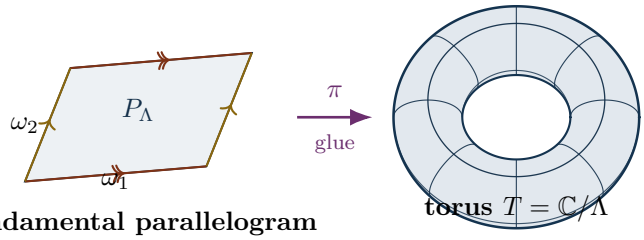
A point of T is an equivalence class

$$[z] = z + \Lambda.$$

The quotient map is

$$\pi : \mathbb{C} \to \mathbb{C}/\Lambda, \qquad z \mapsto [z].$$

Because the transition maps are translations, the complex coordinate z on $\mathbb{C}$ gives local holomorphic coordinates on the quotient. Thus $\mathbb{C}/\Lambda$ is a compact Riemann surface.



3.3 The Group Law

The complex plane is an abelian group under addition. Since Λ is a subgroup, the quotient inherits an abelian group law:

$$[u] + [v] = [u + v].$$

Proposition. *The formula*

$$[u] + [v] = [u + v]$$

is well-defined on $\mathbb{C}/\Lambda$.

Proof. Suppose

$$u' \equiv u \pmod{\Lambda}, \qquad v' \equiv v \pmod{\Lambda}.$$

Then

$$u' = u + \lambda_1, \qquad v' = v + \lambda_2$$

for some $\lambda_1, \lambda_2 \in \Lambda$. Hence

$$u' + v' = u + v + (\lambda_1 + \lambda_2).$$

Since Λ is a subgroup, $\lambda_1 + \lambda_2 \in \Lambda$. Therefore

$$[u' + v'] = [u + v].$$

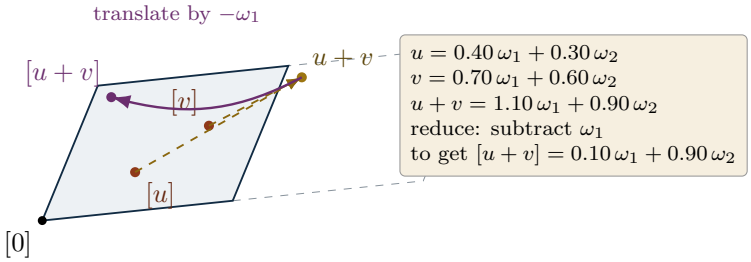
So the operation does not depend on the representatives. □

The identity element is

$$0_T = [0],$$

and the inverse of $[z]$ is

$$-[z] = [-z].$$



3.4 The Torus as a Product of Circles

The abstract group law $[u]+[v]=[u+v]$ has a concrete geometric realization that makes abelianness transparent.

The circle group. The unit circle

$$S^1 = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\}$$

is an abelian group under *complex multiplication*:

$$e^{i\alpha} \cdot e^{i\beta} = e^{i(\alpha+\beta)}.$$

Commutativity holds because $\alpha + \beta = \beta + \alpha$ in $\mathbb{R}$. The exponential map

$$\exp_{2\pi} : \mathbb{R} \to S^1, \qquad t \mapsto e^{2\pi i t},$$

is a surjective group homomorphism with $\ker \exp_{2\pi} = \mathbb{Z}$. By the first isomorphism theorem,

$$\mathbb{R}/\mathbb{Z} \;\cong\; S^1.$$

Thus the circle is the real line *folded into itself*: each unit interval $[n, n+1)$ is wrapped onto S^1 by angle $\theta = 2\pi t$, and the group law becomes angle addition modulo 2π.

Proposition (Torus $\cong$ product of circles)**.** *Let $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$. The map*

$$\Psi : \mathbb{C}/\Lambda \longrightarrow S^1 \times S^1, \qquad [s\,\omega_1 + t\,\omega_2] \longmapsto \big(e^{2\pi i s},\, e^{2\pi i t}\big)$$

is an isomorphism of topological groups.

Proof. Because ω_1, ω_2 are linearly independent over $\mathbb{R}$, every $z \in \mathbb{C}$ has a unique decomposition $z = s\omega_1 + t\omega_2$ with $s, t \in \mathbb{R}$. The real-linear map

$$\phi : \mathbb{R}^2 \to \mathbb{C}, \qquad (s,t) \mapsto s\omega_1 + t\omega_2$$

is an isomorphism of real vector spaces and sends $\mathbb{Z}^2$ onto Λ, so it induces an isomorphism of abelian groups $\mathbb{R}^2/\mathbb{Z}^2 \xrightarrow{\sim} \mathbb{C}/\Lambda$. Composing with the coordinate-wise exponential

$$\mathbb{R}^2/\mathbb{Z}^2 = (\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/\mathbb{Z}) \xrightarrow{(t_1,t_2) \mapsto (e^{2\pi i t_1}, e^{2\pi i t_2})} S^1 \times S^1$$

defines Ψ.

Group homomorphism. Write $[z_k] = [s_k\omega_1 + t_k\omega_2]$ for $k = 1, 2$. Then

$$\begin{aligned}
\Psi\big([z_1] + [z_2]\big) &= \Psi\big([(s_1{+}s_2)\omega_1 + (t_1{+}t_2)\omega_2]\big) \\
&= \big(e^{2\pi i(s_1+s_2)},\; e^{2\pi i(t_1+t_2)}\big) \\
&= \big(e^{2\pi i s_1}\cdot e^{2\pi i s_2},\; e^{2\pi i t_1}\cdot e^{2\pi i t_2}\big) = \Psi([z_1]) \cdot \Psi([z_2]).
\end{aligned}$$

Injectivity. $\Psi([z]) = (1,1)$ forces $e^{2\pi i s} = e^{2\pi i t} = 1$, so $s, t \in \mathbb{Z}$, hence $z = s\omega_1 + t\omega_2 \in \Lambda$ and $[z] = [0]$.

Surjectivity. Given any $(e^{2\pi i s}, e^{2\pi i t}) \in S^1 \times S^1$, the class $[s\omega_1 + t\omega_2]$ maps to it.

Topological isomorphism. Both $\mathbb{C}/\Lambda$ and $S^1 \times S^1$ are compact Hausdorff spaces, and Ψ is continuous (the exponential is continuous), so a continuous bijection between compact Hausdorff spaces is automatically a homeomorphism. $\square$

Corollary. *$T = \mathbb{C}/\Lambda$ is an abelian group.*

Proof. Since complex multiplication is commutative ($wz = zw$ for all $w, z \in \mathbb{C}$), the circle S^1 is an abelian group. A direct product of abelian groups is abelian, so $S^1 \times S^1$ is abelian. The isomorphism Ψ therefore forces $\mathbb{C}/\Lambda$ to be abelian. □

Why commutativity is obvious from the circle picture. A class $[z] = [s\omega_1 + t\omega_2]$ corresponds to the pair of points $(e^{2\pi i s}, e^{2\pi i t})$ on two copies of the unit circle: one tracking position along the ω_1-direction, the other along ω_2. Adding two classes means *rotating each circle independently*:

$$(e^{2\pi i s_1}, e^{2\pi i t_1}) \cdot (e^{2\pi i s_2}, e^{2\pi i t_2}) = (e^{2\pi i (s_1+s_2)},\ e^{2\pi i (t_1+t_2)}).$$

Since rotation by α then by β is the same as rotation by β then by α (addition of real numbers commutes), the group law is automatically commutative. The two copies of S^1 are the two independent "winding directions" on the torus.

3.5 Genus One

Topologically, the quotient of a parallelogram by identifying opposite sides is a torus. The midterm computation gives

$$\chi(T) = 0.$$

For a compact connected Riemann surface of genus g,

$$\chi = 2 - 2g.$$

Thus

$$0 = 2 - 2g, \qquad g = 1.$$

This genus-one fact is crucial. It is why Riemann–Roch has the especially simple form used below.

4 Elliptic Functions as Meromorphic Functions on the Torus

4.1 Periodic Functions

A meromorphic function f on $\mathbb{C}$ is called Λ-periodic, or *elliptic with respect to* Λ, if

$$f(z + \lambda) = f(z) \qquad \text{for all } \lambda \in \Lambda.$$

Such a function descends to a meromorphic function on $T = \mathbb{C}/\Lambda$, because the value of $f(z)$ depends only on the class $[z]$.

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{f} & \mathbb{CP}^1 \\ \downarrow & & \\ \mathbb{C}/\Lambda & & \end{array}$$

Conversely, every meromorphic function on T pulls back to a Λ-periodic meromorphic function on $\mathbb{C}$.

4.2 Why There Are No Nonconstant Holomorphic Functions

The torus T is compact. Therefore every holomorphic function

$$T \to \mathbb{C}$$

is constant by the maximum modulus principle. Equivalently, every Λ-periodic entire function on $\mathbb{C}$ is constant.

Thus nonconstant functions on T must be meromorphic and must have poles. The first question is:

> How small can the pole divisor of a nonconstant meromorphic function on a torus be?

Riemann–Roch answers this.

5 Divisors, Principal Divisors, and Logarithmic Derivatives

5.1 Orders

Let X be a Riemann surface and let f be a nonzero meromorphic function. In a local coordinate t centered at p, we can write

$$f(t) = t^m u(t),$$

where u is holomorphic and $u(0) \neq 0$. The integer m is the order of f at p:

$$\operatorname{ord}_p(f) = m.$$

- If $m > 0$, then f has a zero of order m at p.
- If $m = 0$, then f is nonzero and finite at p.
- If $m < 0$, then f has a pole of order $-m$ at p.

5.2 Divisors

The divisor of f is

$$(f) = \sum_{p \in X} \operatorname{ord}_p(f)\, p.$$

Only finitely many coefficients are nonzero when X is compact.

The degree of a divisor

$$D = \sum_p n_p p$$

is

$$\deg D = \sum_p n_p.$$

A divisor of the form (f) is called *principal*.

5.3 Principal Divisors Have Degree Zero

The final exam proved the following result.

Proposition. *If X is compact and f is a nonzero meromorphic function on X, then*

$$\deg(f) = 0.$$

Proof. Consider the logarithmic derivative

$$\frac{df}{f}.$$

Near p, write

$$f = t^m u(t), \qquad u(0) \neq 0.$$

Then

$$\frac{df}{f} = \frac{d(t^m u)}{t^m u} = m\frac{dt}{t} + \frac{du}{u}.$$

The second term is holomorphic at p. Therefore

$$\operatorname{Res}_p\left(\frac{df}{f}\right) = m = \operatorname{ord}_p(f).$$

By the residue theorem on a compact Riemann surface,

$$\sum_p \operatorname{Res}_p\left(\frac{df}{f}\right) = 0.$$

Hence

$$\sum_p \operatorname{ord}_p(f) = 0,$$

which is exactly

$$\deg(f) = 0.$$

□

What degree zero means. A meromorphic function on a compact Riemann surface has as many zeros as poles, counted with multiplicity. This is the divisor version of the argument principle.

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6 Riemann–Roch on a Genus-One Surface

6.1 The Spaces $L(D)$

For a divisor D on a compact Riemann surface X, define

$$L(D) = \{f \in \mathcal{M}(X) : (f) + D \geq 0\} \cup \{0\}.$$

The condition $(f) + D \geq 0$ means that the poles of f are bounded by the positive part of D, and that f has required zeros where D has negative coefficients.

For example, if $D = nO$ with $n \geq 0$, then

$$L(nO)$$

is the space of meromorphic functions whose only possible pole is at O and whose pole order at O is at most n.

6.2 The Canonical Divisor of a Torus

On $T = \mathbb{C}/\Lambda$, the differential dz is translation-invariant:

$$d(z + \lambda) = dz.$$

Therefore dz descends to a nowhere-vanishing holomorphic 1-form on the quotient torus. Hence its divisor is zero:

$$(dz) = 0.$$

So the canonical divisor is linearly equivalent to 0.

6.3 Riemann–Roch

For a compact Riemann surface of genus g, Riemann–Roch says

$$\ell(D) - \ell(K - D) = \deg D + 1 - g,$$

where

$$\ell(D) = \dim_{\mathbb{C}} L(D)$$

and K is a canonical divisor.

On a torus, $g = 1$ and $K \sim 0$. Therefore

$$\ell(D) - \ell(-D) = \deg D.$$

For $D = nO$ with $n > 0$, the divisor $-nO$ has negative degree. Thus

$$L(-nO) = 0,$$

because a nonzero meromorphic function cannot have a divisor of total degree positive enough to offset a negative-degree condition. Hence

$$\ell(nO) = n \qquad (n \geq 1).$$

6.4 Consequences

For $n = 1$,

$$\ell(O) = 1.$$

Thus

$$L(O) = \mathbb{C}.$$

There is no nonconstant meromorphic function on a torus with only a simple pole at one point.

For $n = 2$,

$$\ell(2O) = 2.$$

The constants account for one dimension, so there is one new nonconstant function with a double pole at O. This function is the Weierstrass $\wp$-function, after a standard normalization.

For $n = 3$,

$$\ell(3O) = 3.$$

A natural basis is

$$1, \quad \wp, \quad \wp'.$$

Divisor	Dimension	Typical basis	Meaning
O	1	1	no simple-pole function
$2O$	2	$1, \wp$	first nonconstant function
$3O$	3	$1, \wp, \wp'$	projective embedding data

Why $\wp$ and $\wp'$ are inevitable. Riemann–Roch says that the first nonconstant meromorphic function on a complex torus appears with a double pole. Its derivative has a triple pole. Thus the first three natural functions are

$$1, \qquad \wp, \qquad \wp'.$$

These are exactly the functions needed to map the torus to a plane cubic.

7 The Weierstrass $\wp$-Function

7.1 Definition and Basic Shape

For a lattice Λ, the Weierstrass function is

$$\wp(z) = \frac{1}{z^2} + \sum_{\lambda \in \Lambda \setminus \{0\}} \left(\frac{1}{(z-\lambda)^2} - \frac{1}{\lambda^2} \right).$$

The subtraction of $1/\lambda^2$ makes the series converge normally on compact subsets away from the lattice. The function is meromorphic on $\mathbb{C}$, with a double pole at each lattice point.

It is Λ-periodic:

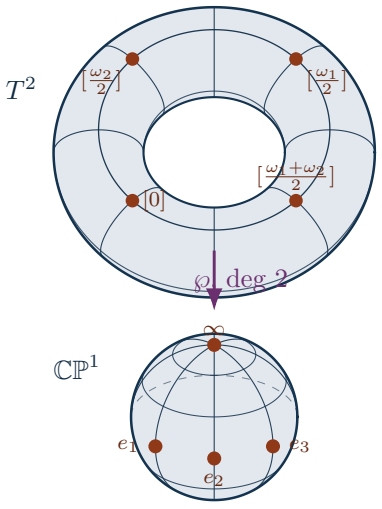
$$\wp(z+\lambda) = \wp(z).$$

It is also even:

$$\wp(-z) = \wp(z).$$

Therefore its derivative is odd:

$$\wp'(-z) = -\wp'(z).$$



The map $\wp : T \to \mathbb{CP}^1$ is a $2:1$ branched cover with 4 branch values: three roots e_1, e_2, e_3 and ∞. Riemann–Hurwitz: $0 = 2{\cdot}2 - 4{\cdot}(2-1)$.

7.2 Laurent Expansions

Near $z = 0$, $\wp$ has the Laurent expansion

$$\wp(z) = \frac{1}{z^2} a_2 z^2 + a_4 z^4 + a_6 z^6 + \cdots .$$

There is no z^{-1} term because $\wp$ has a double pole with zero residue. There are no odd powers because $\wp$ is even.

Differentiating gives

$$\wp'(z) = -\frac{2}{z^3} 2a_2 z 4a_4 z^3 6a_6 z^5 + \cdots .$$

Thus $\wp'$ has a triple pole at the origin and is odd.

7.3 The Differential Equation

The final exam derived the algebraic relation between $\wp$ and $\wp'$. Let us spell out the computation.

Start with

$$\wp(z) = z^{-2} + a_2 z^2 + a_4 z^4 + O(z^6),$$

so

$$\wp'(z) = -2z^{-3} + 2a_2 z + 4a_4 z^3 + O(z^5).$$

Then

$$(\wp')^2 = 4z^{-6} - 8a_2 z^{-2} - 16a_4 + O(z^2),$$

and

$$4\wp^3 = 4z^{-6} + 12a_2 z^{-2} + 12a_4 + O(z^2).$$

Hence

$$(\wp')^2 - 4\wp^3 = -20a_2 z^{-2} - 28a_4 + O(z^2).$$

Choose constants

$$g_2 = 20a_2, \qquad g_3 = 28a_4.$$

Then

$$F(z) = (\wp')^2 - 4\wp^3 + g_2\wp + g_3$$

has no pole at $z = 0$. Since F is elliptic, its only possible poles are at lattice points. By periodicity the same cancellation occurs at every lattice point. Therefore F is holomorphic on the compact torus $\mathbb{C}/\Lambda$, so it is constant.

The expansion also shows that the constant is 0. Therefore

$$\boxed{(\wp')^2 = 4\wp^3 - g_2\wp - g_3.}$$

The birth of the cubic. The equation

$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3$$

is not guessed. It is forced because a carefully chosen combination of $\wp$ and $\wp'$ has no poles, hence must be constant on the compact torus.

7.4 Minimality

The same final-exam argument shows that $\wp'$ has degree 2 over $\mathbb{C}(\wp)$. The relation above says $\wp'$ satisfies

$$T^2 - (4\wp^3 - g_2\wp - g_3) = 0.$$

It remains to see that $\wp' \notin \mathbb{C}(\wp)$.

If $\wp' = R(\wp)$ for a rational function R, then $R(\wp)$ would be even, since $\wp$ is even. But $\wp'$ is odd and nonzero. This is impossible. Thus the quadratic relation is minimal.

8 The $\wp$-Embedding of the Torus

8.1 The Projective Map

Define

$$\Phi : T = \mathbb{C}/\Lambda \longrightarrow \mathbb{CP}^2$$

by

$$\Phi([z]) = [\wp(z) : \wp'(z) : 1]$$

for $z \notin \Lambda$.

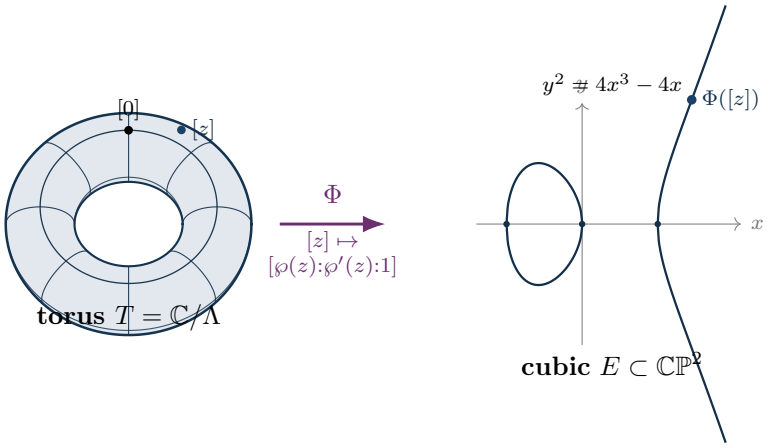
Because $\wp$ and $\wp'$ are periodic, this is well-defined on the quotient. Because they satisfy the differential equation, the image lies on

$$E: \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

Indeed, in the affine chart $Z = 1$, the equation becomes

$$y^2 = 4x^3 - g_2x - g_3,$$

and substituting $x = \wp(z)$, $y = \wp'(z)$ gives exactly the differential equation.



8.2 Extension at the Origin

At $z = 0$, both $\wp$ and $\wp'$ have poles, so the affine formula is not meaningful. But the projective formula has a limit.

Near 0,

$$\wp(z) = z^{-2} + O(z^2), \qquad \wp'(z) = -2z^{-3} + O(z).$$

Thus

$$[\wp(z) : \wp'(z) : 1] = [z^3\wp(z) : z^3\wp'(z) : z^3].$$

As $z \to 0$,

$$z^3\wp(z) \to 0, \qquad z^3\wp'(z) \to -2, \qquad z^3 \to 0.$$

Therefore

$$\lim_{z\to 0}[\wp(z) : \wp'(z) : 1] = [0 : -2 : 0] = [0 : 1 : 0].$$

So we define

$$\Phi(0) = [0 : 1 : 0] = O.$$

Proposition. *The map*

$$\Phi : T \to E, \qquad \Phi(z) = [\wp(z) : \wp'(z) : 1],$$

with $\Phi(0) = O = [0 : 1 : 0]$*, is a holomorphic map from the torus to the projective cubic* E*.*

8.3 The Point at Infinity

The projective cubic is

$$Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

Set $Z = 0$. Then the equation becomes

$$0 = 4X^3.$$

Hence

$$X = 0.$$

The only projective point with $X = 0$ and $Z = 0$ is

$$[0 : 1 : 0].$$

Thus $O = [0 : 1 : 0]$ is the unique point at infinity on E.

8.4 Why This Map Is the Right Embedding

The functions $1, \wp, \wp'$ form a basis of $L(3O)$. The associated projective map is

$$z \longmapsto [1 : \wp(z) : \wp'(z)]$$

up to a reordering of coordinates. We use

$$[\wp(z) : \wp'(z) : 1]$$

so that the affine chart $Z = 1$ has coordinates $x = \wp(z)$, $y = \wp'(z)$.

Riemann–Roch explains why $L(3O)$ is large enough:

$$\ell(3O) = 3.$$

Three independent sections give a map to $\mathbb{CP}^2$. The single algebraic relation among them is the cubic equation.

8.5 Injectivity, Briefly

One can show:

Lemma. *If $\wp(z) = \wp(w)$, then*

$$w \equiv z \quad \text{or} \quad w \equiv -z \pmod{\Lambda}.$$

This reflects the fact that the map

$$T \to \mathbb{CP}^1, \qquad z \mapsto \wp(z)$$

has degree 2, and its two generic fibers are $\{z, -z\}$.

If also

$$\wp'(z) = \wp'(w),$$

then the possibility $w \equiv -z$ forces

$$\wp'(z) = \wp'(-z) = -\wp'(z),$$

so $\wp'(z) = 0$. At such a branch point, $z \equiv -z$, meaning

$$2z \in \Lambda.$$

Thus the two alternatives coincide on the torus. Therefore Φ is injective.

Remark. *The full proof that Φ is a biholomorphism from T onto the smooth cubic E can be obtained by combining injectivity, compactness, and the fact that a nonconstant holomorphic map between compact Riemann surfaces is a finite branched covering. Since T is compact and E is Hausdorff, a continuous bijective holomorphic map $T \to E$ is a biholomorphism.*

9 Inverses Become Reflection Across the x-Axis

The inverse on the torus is

$$z \mapsto -z.$$

Because

$$\wp(-z) = \wp(z), \qquad \wp'(-z) = -\wp'(z),$$

we get

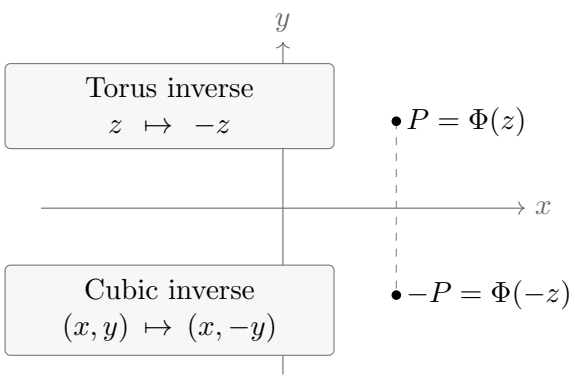
$$\Phi(-z) = [\wp(-z) : \wp'(-z) : 1] = [\wp(z) : -\wp'(z) : 1].$$

In affine coordinates, if

$$P = (x, y) = (\wp(z), \wp'(z)),$$

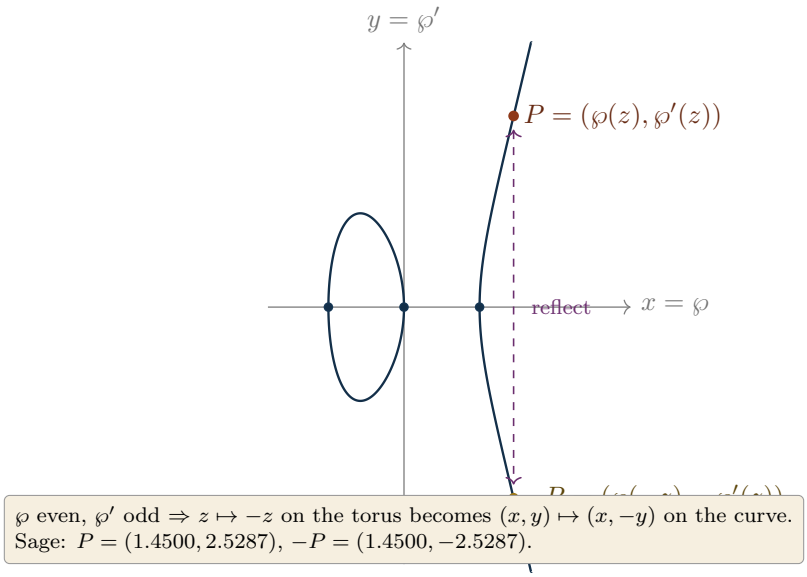
then

$$-P = (x, -y).$$



Why reflection appears. Reflection across the x-axis is not an extra convention. It is exactly the torus inverse $z \mapsto -z$ written in the coordinates

$$x = \wp(z), \qquad y = \wp'(z).$$



$\wp$ even, $\wp'$ odd $\Rightarrow z \mapsto -z$ on the torus becomes $(x, y) \mapsto (x, -y)$ on the curve.
Sage: $P = (1.4500, 2.5287)$, $-P = (1.4500, -2.5287)$.

10 Abel's Sum Law on a Torus

10.1 Statement

The central divisor fact is the genus-one form of Abel's theorem.

Theorem (Abel sum law for elliptic functions)**.** *Let F be a nonzero meromorphic function on*

$$T = \mathbb{C}/\Lambda.$$

Write its divisor as

$$(F) = \sum_j n_j [z_j],$$

where $z_j \in \mathbb{C}$ are representatives of points on T. Then

$$\sum_j n_j = 0$$

and

$$\sum_j n_j z_j \in \Lambda.$$

Equivalently,

$$\sum_j n_j [z_j] = 0$$

in the group law of T.

The first equation is the degree-zero property of principal divisors. The second equation says that zeros and poles also balance in the torus group law.

10.2 Proof of the Degree Condition

Pull F back to a Λ-periodic meromorphic function on $\mathbb{C}$. Choose a fundamental parallelogram whose boundary contains no zeros or poles of F. The logarithmic derivative

$$\frac{F'}{F}\,dz$$

is periodic. By the residue theorem,

$$\frac{1}{2\pi i}\int_{\partial P_\Lambda}\frac{F'}{F}\,dz = \sum_j n_j.$$

Opposite sides of the parallelogram cancel because F'/F is periodic. Therefore the boundary integral is 0. Hence

$$\sum_j n_j = 0.$$

10.3 Proof of the Abel-Sum Condition

Now integrate

$$z\frac{F'}{F}\,dz$$

around the same fundamental parallelogram. Its residues are

$$\mathrm{Res}_{z_j}\left(z\frac{F'}{F}\,dz\right) = n_j z_j.$$

Therefore

$$\frac{1}{2\pi i}\int_{\partial P_\Lambda} z\frac{F'}{F}\,dz = \sum_j n_j z_j.$$

Unlike F'/F, the function zF'/F is not periodic because of the factor z. The opposite sides no longer cancel exactly. The failure of cancellation is a period.

Let the parallelogram have sides ω_1 and ω_2. Since

$$\frac{F'}{F}$$

is periodic, the boundary integral reduces to

$$\omega_1\int_{\text{one }\omega_2\text{-side}}\frac{F'}{F}\,dz - \omega_2\int_{\text{one }\omega_1\text{-side}}\frac{F'}{F}\,dz.$$

Each side integral is a change of a branch of $\log F$. Since $F(z+\omega_k) = F(z)$, that change is an integer multiple of $2\pi i$. Thus

$$\int_{\partial P_\Lambda} z\frac{F'}{F}\,dz \in 2\pi i\,\Lambda.$$

Dividing by $2\pi i$ gives

$$\sum_j n_j z_j \in \Lambda.$$

Meaning of Abel's sum law. A principal divisor on a torus is not only degree zero. Its weighted sum of points is also zero in the group law of the torus.

$$(F) = \sum n_j p_j \quad \Longrightarrow \quad \sum n_j p_j = 0 \quad \text{in } T.$$

11 Lines on the Cubic Give Principal Divisors

11.1 A Projective Line

A projective line in $\mathbb{CP}^2$ is given by a homogeneous linear equation

$$\ell(X, Y, Z) = aX + bY + cZ.$$

Restricting ℓ to the cubic E gives a meromorphic function only after we divide by another homogeneous function of the same degree. The standard comparison function is Z, the line at infinity.

Thus

$$\frac{\ell}{Z}$$

is a meromorphic function on E.

11.2 Intersection Divisors

If the line $L = \{\ell = 0\}$ meets E in points

$$P, Q, R,$$

counted with multiplicity, then

$$\operatorname{div}_0(\ell|_E) = P + Q + R.$$

This is Bézout's theorem in the special case of a line and a cubic: a degree-one curve and a degree-three curve meet in 3 points, counted with intersection multiplicity.

11.3 The Divisor of the Line at Infinity

The line at infinity is

$$Z = 0.$$

On

$$E: \quad Y^2 Z = 4X^3 - g_2 X Z^2 - g_3 Z^3,$$

we already saw that $Z = 0$ meets E only at

$$O = [0 : 1 : 0].$$

The multiplicity is 3.

Here is a local check. Work in the chart $Y = 1$ near O. Put

$$u = \frac{X}{Y}, \qquad v = \frac{Z}{Y}.$$

Then O is $(u, v) = (0, 0)$, and the cubic equation becomes

$$v = 4u^3 - g_2uv^2 - g_3v^3.$$

Near $(0, 0)$, this equation implies

$$v = 4u^3 + \text{higher-order terms}.$$

Thus u is a local parameter and

$$Z/Y = v$$

vanishes to order 3 at O. Therefore

$$\operatorname{div}_0(Z|_E) = 3O.$$

11.4 The Principal Divisor Associated to a Line

Since ℓ/Z is a meromorphic function on E, its divisor is principal:

$$\left(\frac{\ell}{Z}\right) = \operatorname{div}_0(\ell|_E) - \operatorname{div}_0(Z|_E).$$

If

$$L \cdot E = P + Q + R,$$

then

$$\boxed{\left(\frac{\ell}{Z}\right) = P + Q + R - 3O.}$$

The line divisor. Every projective line cuts the cubic in a divisor linearly equivalent to $3O$:

$$P + Q + R \sim 3O.$$

Equivalently,

$$P + Q + R - 3O$$

is principal.

---

12 Transporting the Torus Group Law to the Cubic

The $\wp$-embedding gives a biholomorphic identification

$$\Phi : T = \mathbb{C}/\Lambda \longrightarrow E.$$

We transport the torus group law to E by declaring

$$P +_E Q = \Phi(u + v)$$

whenever

$$P = \Phi(u), \qquad Q = \Phi(v).$$

The identity is

$$O = \Phi(0) = [0 : 1 : 0].$$

The inverse is

$$-\Phi(u) = \Phi(-u).$$

In affine coordinates this is

$$-(x, y) = (x, -y).$$

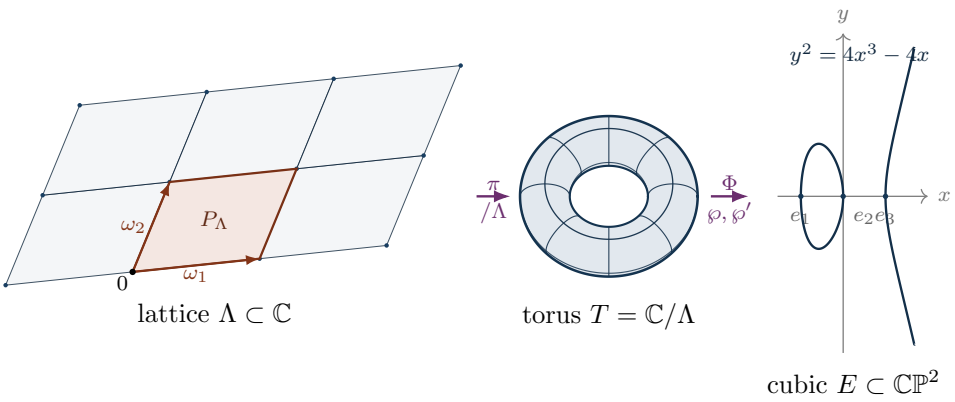
The question is:

Why does this transported group law coincide with the chord–tangent law?

The answer is: because line divisors are principal and principal divisors have Abel sum zero.

13 Pictorial Dictionary: Torus Addition vs. Chord–Tangent

The map $\Phi : \mathbb{C}/\Lambda \to E$ is a group isomorphism. Each addition on the left corresponds to a chord–tangent construction on the right. The three figures below display this dictionary for the three structurally distinct cases: adding the identity, adding two distinct nonopposite points, and doubling a single point.



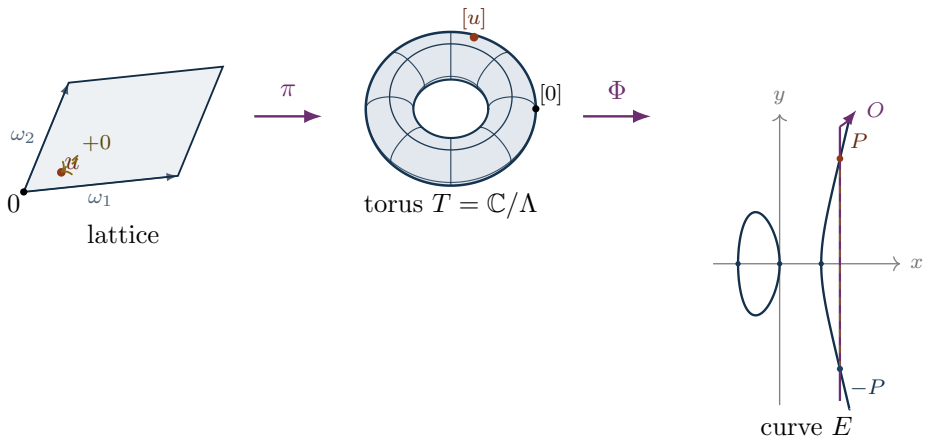
lattice $\Lambda \subset \mathbb{C}$ torus $T = \mathbb{C}/\Lambda$ cubic $E \subset \mathbb{CP}^2$

Throughout, the rust colour marks the input points, plum marks the auxiliary line (chord, tangent, or vertical), teal marks the third intersection R, and gold marks the final result obtained by reflecting R across the x-axis.

All coordinates in the figures below are computed in `sagemath/11_correspondence.sage`. Each marker $[w] = a\,\omega_1 + b\,\omega_2$ is shown in three locations: its position in the fundamental parallelogram P_Λ; its embedded 3D position on the torus via $u_t = 2\pi b$, $v_t = 2\pi a$ (small-circle / big-circle angles, then orthographic projection at $\theta = 60°$); and the corresponding chord/tangent picture on $E : y^2 = 4x^3 - 4x$. Lattice coefficients (a, b) are kept small enough that every embedded marker passes the visibility check $-\cos u_t \sin v_t \cos\theta + \sin u_t \sin\theta > 0$.

13.1 Identity element: $P + O = P$

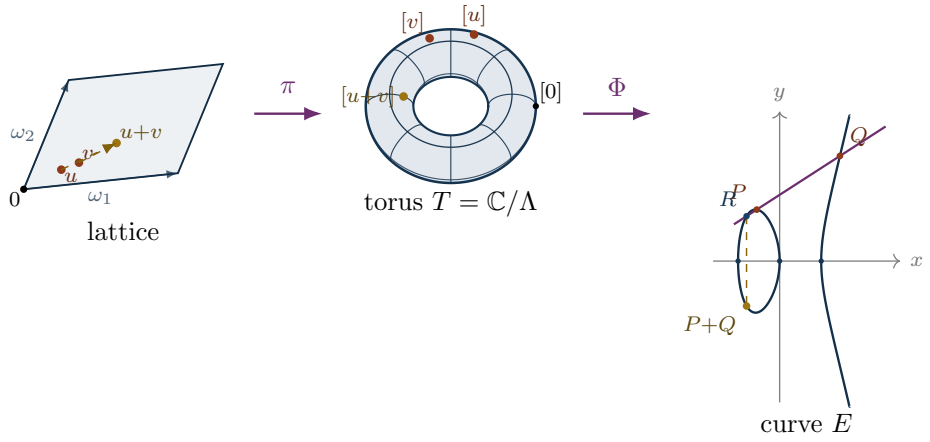
On the torus, adding the identity 0 is the trivial translation $[u] \mapsto [u] + 0 = [u]$. Under Φ, the identity 0 maps to the point at infinity O, and the line through P and O is the vertical line through P; its third intersection with the cubic is $-P$, and reflecting $-P$ across the x-axis recovers P.



Reading the figure. On the lattice (left) the self-loop at u records the trivial translation $u + 0 = u$. The same points $[u], [0]$ appear at their embedded torus positions (middle) via $(v_t, u_t) = (2\pi a, 2\pi b)$. On the curve (right) the chord $\overline{PO}$ is vertical; it meets E at P, $-P$, and O, so the chord-tangent rule says $P + O + (-P) = O$, i.e. $P + O = P$.

13.2 Two distinct points: $P + Q$

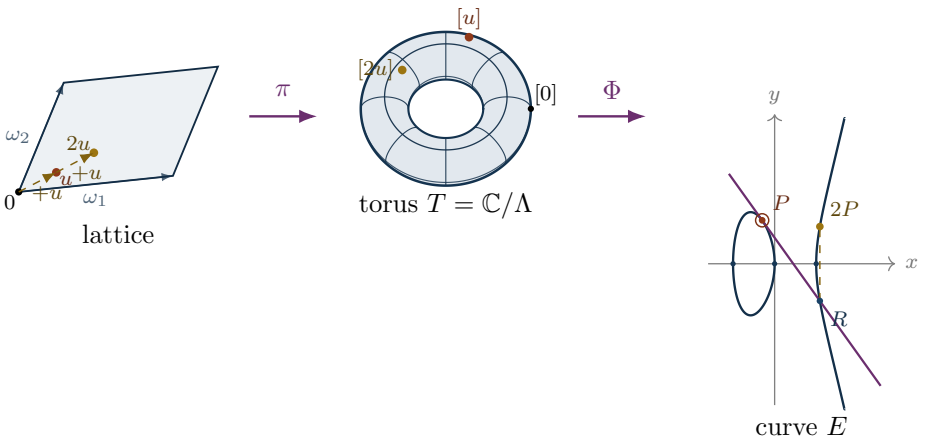
On the torus, addition is just vector addition modulo Λ: from $[u]$ translate by v to reach the lifted sum $u + v$, then reduce modulo Λ back into P_Λ. On the cubic, draw the chord through P and Q; it meets E at a third point R, and reflecting R across the x-axis yields $P + Q = -R$.



Reading the figure. The two dashed gold arrows on the parallelogram show $u \to u+v$ and $v \to u+v$; under π those two translates land at the same point $[u + v]$ on the torus. The chord on E then meets the cubic at a third point R; reflecting gives $P + Q = -R$. Numerically (from Sage): $P = (-0.5500, 1.2387)$, $Q = (1.4500, 2.5287)$, $R = (-0.7960, 1.0801)$, $P + Q = (-0.7960, -1.0801)$.

13.3 Doubling a point: $2P$

The doubling case is the limit of the chord rule as $Q \to P$: the chord becomes the tangent line at P with intersection multiplicity 2 at P. On the torus, doubling $[u]$ is just $[u] + [u] = [2u]$, the ordinary vector $2u$ reduced modulo Λ.



Reading the figure. Two consecutive $+u$ translations on the parallelogram give the lifted vector $2u$; under π this lands at $[2u]$ on the torus, marked in gold. Under Φ the marker $[u]$ maps to P, and the chord limit (i.e. tangent at P) hits the cubic at R; reflecting gives $2P = -R$. Numerically: $P = (-0.3000, 1.0450)$, tangent slope $m = \frac{12x_P^2 - g_2}{2y_P}$, $R = (1.0880, -0.8943)$, $2P = (1.0880, 0.8943)$.

Remark. *The three figures share the same colour code: rust for the input(s), plum for the auxiliary line (vertical / chord / tangent), teal for the residual intersection R, and gold for the final result. The single visual rule shared by every case is:*

$$\boxed{\textit{points + line on } E \xrightarrow{\textit{Vieta}} R \xrightarrow{y \mapsto -y} R + (\textit{stuff}) = O, \ \textit{so result} = -R.}$$

14 The Chord Law: Distinct Points

14.1 Setup

Let $P, Q \in E$ be two distinct finite points with $Q \neq -P$. Let L be the line through P and Q. Since L is not vertical in this case, it meets the cubic in a third finite point R.

Write

$$P = \Phi(u), \qquad Q = \Phi(v), \qquad R = \Phi(w).$$

The intersection divisor is

$$L \cdot E = P + Q + R.$$

Therefore

$$\left(\frac{\ell}{Z}\right) = P + Q + R - 3O.$$

14.2 Apply Abel's Sum Law

Pull this principal divisor back to the torus. Since $O = \Phi(0)$, Abel's sum law gives

$$u + v + w - 3 \cdot 0 \equiv 0 \pmod{\Lambda}.$$

Hence

$$w \equiv -(u + v) \pmod{\Lambda}.$$

Therefore

$$R = \Phi(w) = \Phi(-(u + v)).$$

But the torus sum of P and Q is

$$P +_E Q = \Phi(u + v).$$

Since inverse on the cubic is reflection,

$$\Phi(u + v) = -\Phi(-(u + v)).$$

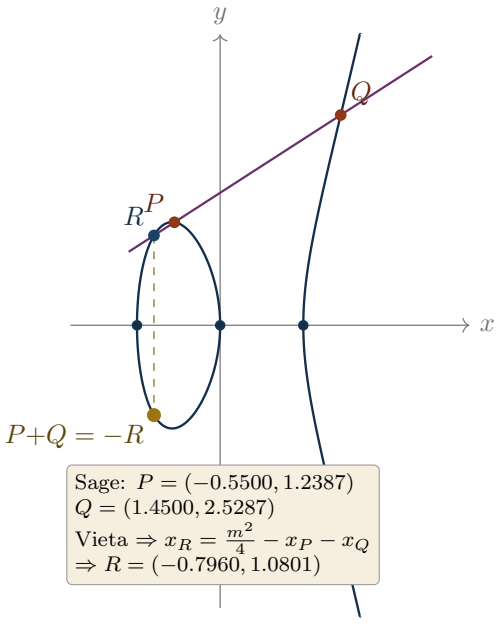
Thus

$$\boxed{P +_E Q = -R.}$$

14.3 Geometric Rule

This is exactly the chord rule:

1. Draw the line through P and Q.
2. Let R be the third intersection with the cubic.
3. Reflect R across the x-axis.
4. The reflected point is $P + Q$.



Why the sign appears. The line gives $P + Q + R = O$, not $P + Q = R$. Therefore

$$P + Q = -R.$$

The final reflection is exactly this minus sign.

---

15 The Tangent Law: Doubling a Point

15.1 Intersection Multiplicity

If $P = Q$, the line through P and Q is replaced by the tangent line at P. This is not a new rule. It is the limiting case of the chord rule as $Q \to P$.

Algebraically, the tangent line has intersection multiplicity 2 at P. If it meets the cubic again at R, then

$$L \cdot E = 2P + R.$$

15.2 Divisor Argument

Write

$$P = \Phi(u), \qquad R = \Phi(w).$$

The line divisor gives

$$\left(\frac{\ell}{Z}\right) = 2P + R - 3O.$$

Abel's sum law gives

$$2u + w \equiv 0 \pmod{\Lambda}.$$

Hence

$$w \equiv -2u \pmod{\Lambda}.$$

Therefore

$$R = \Phi(-2u),$$

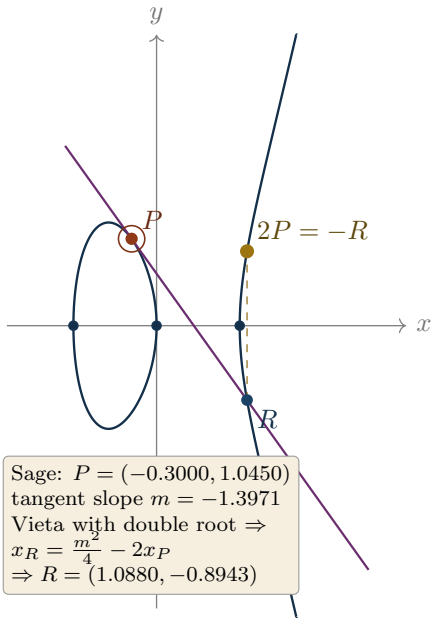
and

$$2P = \Phi(2u) = -R.$$

15.3 Geometric Rule

To compute $2P$:

1. Draw the tangent line to E at P.
2. Let R be the third intersection with E.
3. Reflect R across the x-axis.
4. The reflected point is $2P$.



16 The Identity, Inverses, and Vertical Lines

16.1 The Identity

The identity element on the torus is 0. Under the embedding,

$$0 \longmapsto O = [0 : 1 : 0].$$

Therefore O is the identity on E.

Geometrically, adding O should do nothing:

$$P + O = P.$$

This is compatible with the line rule. A line through a finite point P and O is the vertical line through P. It also passes through $-P$. Thus

$$P + O + (-P) = O,$$

so

$$P + O = P.$$

16.2 Vertical Lines

Let

$$P = (x, y).$$

Then

$$-P = (x, -y).$$

The vertical line

$$X = xZ$$

passes through

$$P, \quad -P, \quad O.$$

The intersection divisor is

$$P + (-P) + O.$$

The line divisor gives

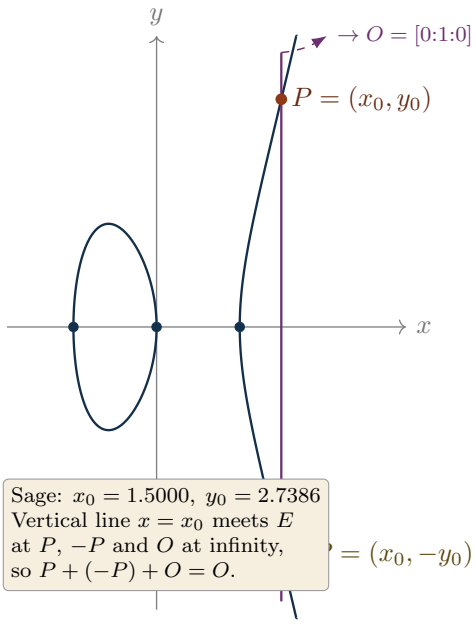
$$P + (-P) + O - 3O = P + (-P) - 2O.$$

Abel's sum law says

$$P + (-P) + O = O$$

in the group law, hence

$$P + (-P) = O.$$



16.3 Two-Torsion Points

If $P = -P$, then

$$2P = O.$$

In affine coordinates this means

$$(x, y) = (x, -y),$$

so

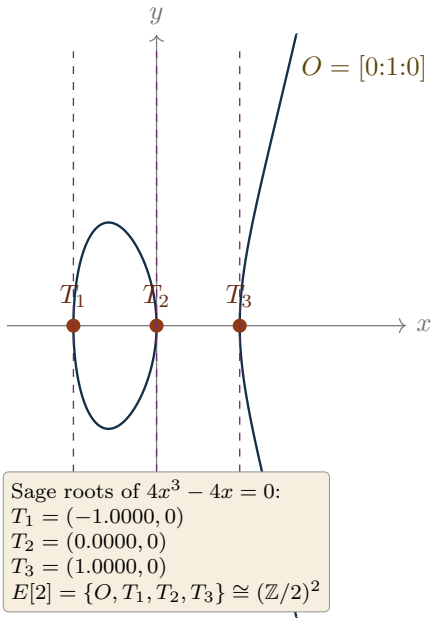
$$y = 0.$$

Thus the nonzero two-torsion points are the points where the cubic crosses the x-axis:

$$4x^3 - g_2 x - g_3 = 0.$$

At such a point, the tangent line is vertical, and the doubling rule gives

$$2P = O.$$



17 Coordinate Formulas for the Group Law

The divisor proof explains why the chord–tangent construction is correct. It is also useful to have explicit formulas.

17.1 Addition of Distinct Noninverse Points

Let

$$P = (x_1, y_1), \qquad Q = (x_2, y_2)$$

be finite points on

$$E: \quad y^2 = 4x^3 - g_2 x - g_3.$$

Assume

$$P \neq Q, \qquad Q \neq -P.$$

Then $x_1 \neq x_2$, and the slope of the line through P and Q is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

The line is

$$y = mx + b, \qquad b = y_1 - mx_1.$$

Substitute this into the cubic:

$$(mx + b)^2 = 4x^3 - g_2 x - g_3.$$

Rearrange:

$$4x^3 - m^2 x^2 - (g_2 + 2mb)x - (g_3 + b^2) = 0.$$

The roots are the x-coordinates of the three intersection points:

$$x_1, \quad x_2, \quad x_R.$$

By Vieta's formula,

$$x_1 + x_2 + x_R = \frac{m^2}{4}.$$

Thus

$$x_R = \frac{m^2}{4} - x_1 - x_2.$$

Since

$$P + Q = -R,$$

the x-coordinate of $P + Q$ is the same:

$$\boxed{x_{P+Q} = \frac{m^2}{4} - x_1 - x_2.}$$

The y-coordinate is the negative of the line value at x_R:

$$\boxed{y_{P+Q} = -(mx_{P+Q} + b).}$$

17.2 Doubling Formula

For

$$P = (x_1, y_1)$$

with $y_1 \neq 0$, the tangent slope is found by implicit differentiation. Starting from

$$y^2 = 4x^3 - g_2x - g_3,$$

differentiate:

$$2y\frac{dy}{dx} = 12x^2 - g_2.$$

Therefore

$$m = \left.\frac{dy}{dx}\right|_P = \frac{12x_1^2 - g_2}{2y_1}.$$

Then the same formulas apply:

$$\boxed{x_{2P} = \frac{m^2}{4} - 2x_1,}$$

and

$$\boxed{y_{2P} = -(mx_{2P} + b), \qquad b = y_1 - mx_1.}$$

17.3 Special Cases

Situation	Result
$P = O$	$P + Q = Q$
$Q = O$	$P + Q = P$
$Q = -P$	$P + Q = O$
$P = Q,\ y_P = 0$	$2P = O$
$P = Q,\ y_P \neq 0$	use tangent formula

18 Analytic Addition Formula for $\wp$

18.1 The Formula

The Weierstrass addition formula says

$$\boxed{\wp(u+v) = -\wp(u) - \wp(v) + \frac{1}{4}\left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}\right)^2}$$

whenever the expression is defined; the exceptional cases follow by limits.

Let

$$P = \Phi(u) = (\wp(u), \wp'(u)), \qquad Q = \Phi(v) = (\wp(v), \wp'(v)).$$

Set

$$x_1 = \wp(u), \quad y_1 = \wp'(u), \qquad x_2 = \wp(v), \quad y_2 = \wp'(v).$$

The slope of the chord is

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\wp'(v) - \wp'(u)}{\wp(v) - \wp(u)}.$$

This equals

$$\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}$$

after multiplying numerator and denominator by -1.

The coordinate formula from the chord computation gives

$$x_{P+Q} = \frac{m^2}{4} - x_1 - x_2.$$

The addition formula gives exactly

$$\wp(u+v) = \frac{m^2}{4} - \wp(u) - \wp(v).$$

Thus

$$x_{P+Q} = \wp(u+v).$$

18.2 Why the y-Coordinate Has the Reflected Sign

The third intersection point is

$$R = \Phi(-(u+v)).$$

Its coordinates are

$$R = (\wp(-(u+v)), \wp'(-(u+v))).$$

Since $\wp$ is even and $\wp'$ is odd,

$$R = (\wp(u+v), -\wp'(u+v)).$$

Therefore

$$-R = (\wp(u+v), \wp'(u+v)).$$

But

$$\Phi(u+v) = (\wp(u+v), \wp'(u+v)).$$

Hence

$$P + Q = \Phi(u+v) = -R.$$

Analytic and geometric agreement. The addition formula for $\wp$ is the analytic version of the chord law. The root-sum calculation gives the same x-coordinate as $\wp(u+v)$, and the oddness of $\wp'$ supplies the reflection.

19 Why the Chord–Tangent Law Is Forced

We can now isolate the exact logical mechanism.

19.1 There Is Already a Group Law

The torus $T = \mathbb{C}/\Lambda$ already has an abelian group law:

$$[u] + [v] = [u + v].$$

There is no choice here. It comes directly from addition in $\mathbb{C}$.

19.2 The $\wp$-Embedding Transfers That Law

The map

$$\Phi : T \to E$$

identifies the torus with the cubic. Therefore it transports the group law:

$$P +_E Q = \Phi(\Phi^{-1}(P) + \Phi^{-1}(Q)).$$

Again, there is no choice if we require Φ to be a group isomorphism.

19.3 Lines Know the Same Group Law

For every projective line L,

$$L \cdot E = P + Q + R$$

and

$$P + Q + R - 3O$$

is principal. Abel's sum law says principal divisors have group sum zero. Therefore

$$P + Q + R = O.$$

Solving for $P + Q$ gives

$$P + Q = -R.$$

19.4 The Result

Theorem. *Under the Weierstrass embedding*

$$\Phi : \mathbb{C}/\Lambda \to E, \qquad z \mapsto [\wp(z) : \wp'(z) : 1],$$

the abelian group law on $\mathbb{C}/\Lambda$ becomes the chord–tangent group law on

$$E : \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

Proof. Let $P = \Phi(u)$ and $Q = \Phi(v)$. If P, Q are distinct and not inverse, let the line through them meet E again at $R = \Phi(w)$. The line divisor gives the principal divisor

$$P + Q + R - 3O.$$

By Abel's sum law,

$$u + v + w \equiv 0 \pmod{\Lambda}.$$

Thus

$$w \equiv -(u + v).$$

Therefore

$$R = \Phi(-(u + v)).$$

Reflecting gives

$$-R = \Phi(u + v),$$

which is the transported torus sum of P and Q. The tangent, vertical-line, and identity cases are the same argument with intersection multiplicities. Hence the transported group law is exactly the chord–tangent law. □

No arbitrary convention. The chord–tangent rule is forced by three facts:

$$\text{torus addition} \quad + \quad \wp\text{-embedding} \quad + \quad \text{principal line divisors.}$$

The reflection step is forced because the line gives

$$P + Q + R = O,$$

so the desired sum is $-R$.

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20 A Complete Worked Symbolic Computation

This section performs the chord computation from beginning to end in coordinates.

20.1 Start with Two Points

Let

$$P = (x_1, y_1), \qquad Q = (x_2, y_2)$$

on

$$y^2 = 4x^3 - g_2 x - g_3.$$

Assume $x_1 \neq x_2$. Define

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \qquad b = y_1 - m x_1.$$

The chord is

$$y = mx + b.$$

20.2 Substitute

Plug into the cubic:

$$(mx + b)^2 = 4x^3 - g_2x - g_3.$$

Move everything to one side:

$$4x^3 - m^2x^2 - (g_2 + 2mb)x - (g_3 + b^2) = 0.$$

Since P and Q lie on both the line and the cubic, x_1 and x_2 are roots. Let the third root be x_3. Then

$$4(x - x_1)(x - x_2)(x - x_3)$$

has the same leading coefficient and the same roots. Comparing the x^2-coefficient:

$$-4(x_1 + x_2 + x_3) = -m^2.$$

Therefore

$$x_3 = \frac{m^2}{4} - x_1 - x_2.$$

20.3 Find the Third Point and Reflect

The third intersection point is

$$R = (x_3, y_3), \qquad y_3 = mx_3 + b.$$

The group sum is

$$P + Q = -R.$$

Since inverse is reflection,

$$-R = (x_3, -y_3).$$

Therefore

$$\boxed{P + Q = \left(\frac{m^2}{4} - x_1 - x_2,\ -m\left(\frac{m^2}{4} - x_1 - x_2\right) - b\right).}$$

20.4 Comparison with the More Common Cubic $y^2 = x^3 + ax + b$

Many textbooks use

$$y^2 = x^3 + ax + b.$$

In that normalization, the addition formula is

$$x_{P+Q} = m^2 - x_1 - x_2.$$

Our cubic is

$$y^2 = 4x^3 - g_2x - g_3.$$

The leading coefficient is 4, so Vieta's formula gives

$$x_{P+Q} = \frac{m^2}{4} - x_1 - x_2.$$

The factor $1/4$ is not mysterious; it comes from the coefficient of x^3.

21 Common Pitfalls and How to Avoid Them

1. **Forgetting the point at infinity.** The cubic group law is not complete unless $O = [0 : 1 : 0]$ is included. Vertical lines meet the cubic at O, and O is the identity.

2. **Saying the third intersection is $P + Q$.** The third intersection is R, but the sum is $-R$. The line gives
$$P + Q + R = O.$$

3. **Ignoring multiplicity.** When $P = Q$, the tangent line counts P twice:
$$L \cdot E = 2P + R.$$
When the tangent has higher contact, the same divisor rule still applies.

4. **Treating reflection as a visual trick.** Reflection is the inverse map on the torus:
$$z \mapsto -z.$$
It becomes
$$(x, y) \mapsto (x, -y)$$
because $\wp$ is even and $\wp'$ is odd.

5. **Using Riemann–Roch without checking genus.** The dimension counts
$$\ell(nO) = n$$
depend on the fact that the torus has genus 1 and $K \sim 0$.

6. **Confusing affine and projective equations.** The affine equation
$$y^2 = 4x^3 - g_2 x - g_3$$
misses the point O. The correct compact curve is
$$Y^2 Z = 4X^3 - g_2 X Z^2 - g_3 Z^3.$$

22 Lecture Summary

The complete chain is:

1. A lattice $\Lambda \subset \mathbb{C}$ gives a compact genus-one Riemann surface
$$T = \mathbb{C}/\Lambda.$$

2. The torus has an intrinsic abelian group law:
$$[u] + [v] = [u + v].$$

3. Riemann–Roch on a genus-one surface says the first nonconstant meromorphic function with one pole has a double pole. This is $\wp$.

4. The functions
$$1, \wp, \wp'$$
span $L(3O)$, so they define a map
$$\Phi : T \to \mathbb{CP}^2.$$

5. The differential equation
$$(\wp')^2 = 4\wp^3 - g_2\wp - g_3$$
shows that the image is the cubic
$$E: \quad Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

6. The point $0 \in T$ maps to
$$O = [0 : 1 : 0].$$

7. The inverse map $z \mapsto -z$ becomes
$$(x, y) \mapsto (x, -y).$$

8. A line L cuts the cubic in
$$P + Q + R.$$
Since $Z = 0$ cuts the cubic in $3O$, the divisor
$$P + Q + R - 3O$$
is principal.

9. Principal divisors on a torus have Abel sum zero. Therefore
$$P + Q + R = O.$$

10. Hence
$$P + Q = -R,$$
which is exactly the chord–tangent law.

Final takeaway. The chord–tangent law is the visible projective shadow of the simple addition law
$$[u] + [v] = [u + v]$$
on the complex torus. The Weierstrass embedding turns addition into line intersection because line intersections give principal divisors, and principal divisors have zero Abel sum.
